Applications of the Divergence Theorem to Gravitational Fields

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Abstract

In this paper, we use the Divergence Theorem to prove both parts of the Newton Shell Theorem. We begin by presenting a simple example to show how the Divergence Theorem works when converting a surface integral into a volume integral. Then we apply the theorem to the gravitational field and show why the inside of a spherical shell has no net gravitational force, and why outside the shell the force acts as if all the mass were at the center. To conclude, we apply these ideas to calculate the gravitational field of the exoplanet Kepler-452b's shell as a real-world example.

1 Introduction

The Divergence Theorem is one of the most prominent results in Vector Calculus. It has many applications to real-world phenomena. More specifically, finding the flux within a region. It takes a surface integral, relating to a closed surface, and maps it to a triple integral with the volume of the object. This has many applications in the realm of physics. For example, finding the flux of an object that has an electrical or magnetic field. In the relation of Gravitational Fields, the Divergence theorem provides a mathematical tool to describe the properties of the Gravitational Field with its Divergence. This is important to physicists and astrophysicists because with this information, we can predict the behavior of the gravitational field. We can also take this information and see how it governs the way our universe moves.

Moreover, the Divergence Theorem makes the complete connection between local and global properties of a vector field. Instead of studying how a field behaves along a curved surface, the Divergence theorem allows us to study how a vector field spreads out or compresses inward. As we mentioned before, this is important to gravitational physics. This idea of measuring how much a vector field compresses or expands outward is the main idea behind the Newton Shell theorem. Historically, Issac Newton proved his idea with geometric intuition and heavy algebra. The vector calculus approach provides a simple, clear solution to Newton's idea. Using the Divergence Theorem to prove this is similar to Gauss' Law within an electric field, which states that the total electric flux through a closed surface depends only on the charge enclosed. Here, the gravitational field depends on the mass enclosed. So if no mass is enclosed inside a spherical region, the net gravitational flux must be zero.

1.1 The Newton Shell Theorem

The Newton Shell Theorem has two parts: first a mass m inside a uniform spherical shell will experience no net gravitational force from the shell. Secondly, the gravitational force on a mass m outside the spherical shell is equivalent to all of the shell's matter concentrated at the center of the spherical shell as seen in [1] and [4].

1.2 The Divergence Theorem

Definition 1. *Let D be a solid region, and S be the boundary surface of E with positive orientation. Let $\vec{F}$ be a vector field whose components have continuous first partial derivatives. Then,*

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_D (\nabla \cdot \vec{F})\, dV$$

We will use this as a fact, proving the Divergence Theorem is beyond the scope of this paper.

1.3 Example Using the Divergence Theorem

Consider the following, from [3]. Compute the flux $\iint \vec{F} \cdot n\, d\vec{S}$ by the Divergence Theorem. Where $\vec{F} = \langle x^2, y^2, z^2 \rangle$ and $S : x^2 + y^2 + z^2 = 1$ (The Unit Sphere).

Solution: We know that $\text{Div}(\vec{F}) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

So we compute:

- $\frac{\partial F_1}{\partial x} = 2x$
- $\frac{\partial F_2}{\partial y} = 2y$
- $\frac{\partial F_3}{\partial z} = 2z$

So, $\text{Div}\vec{F} = 2x + 2y + 2z$. Now we proceed by the definition of the Divergence Theorem. Where E is the unit ball (solid sphere) and S is the unit sphere (surface).

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \nabla \cdot \vec{F}\, dV$$

We have:

$$\iiint_E (2x + 2y + 2z)\, dV$$

By splitting the integral into three parts, we notice symmetry; this is symmetric about the origin, and thus there is a -x for x. So we can deduce that:

$$\iiint_E x\, dV = 0$$

This will hold for y and z as well. Therefore, the flux is 0 because $0 + 0 + 0 = 0$.

2 Problem Statement

In this paper, we will use the Divergence Theorem to prove both parts of the Newton Shell Theorem. More specifically,

1. A point mass located inside the uniform spherical shell will experience no net gravitational force from the shell.
2. A point mass outside the uniform spherical shell will experience a gravitational force equivalent to the shell's total mass concentrated at its center.

This route will highlight the fundamental relationship between flux and divergence, avoiding the cumbersome computations of integrating over the direct volume, which isn't always feasible.

3 Derivation

We know for a gravitational field of a point mass M is

$$\vec{g}(\vec{r}) = -GM\frac{\vec{r}}{r^3}$$

Now we will consider a spherical shell which we can think of this as a planet. Using a similar approach of Gauss' Law to find the flux of an electric field, we will study the flux of $\vec{g}$ through a spherical surface. Any region with no mass satisfies:

$$\nabla \cdot \vec{g} = 0$$

Case 1: Region $r > R$ (Outside the Sphere)

In this region we have our Gaussian Surface encloses the entire mass of our "planet". But we notice that this region between our Gaussian Surface and planet contains no mass. By the fact that $\nabla \cdot \vec{g} = 0$ everywhere and the use of the divergence theorem this implies:

$$\iint_{R_{outside}} \vec{g} \cdot d\vec{S} = \iiint_V (\nabla \cdot \vec{g})\, dV$$

Since there is no mass:

$$\nabla \cdot \vec{g} = 0 \implies \iiint_V (\nabla \cdot \vec{g})\, dV = 0$$

We note that our shell is perfectly symmetric we can simplify the flux to:

$$\iint_{R_{outside}} \vec{g} \cdot d\vec{S} = g(r) \cdot 4\pi r^2$$

Gauss' Law for gravitational fields states that the flux must be equal to the mass enclosed by the surface which is not zero. So,

$$\iint_{R_{outside}} \vec{g} \cdot d\vec{S} = -4\pi GM$$

By setting the two equations together and solving for $g(r)$ we get:

$$g(r) \cdot 4\pi r^2 = -4\pi GM \implies g(r) = -\frac{GM}{r^2}$$

This is exactly what we wanted to show for the second part of the Newton shell theorem. That any point mass outside the shell will experience a gravitational force equivalent to the shell's total mass concentrated at its center.

Case 2: Region $r < R$ (Inside the Sphere)

We consider a Gaussian surface of radius r which is located in the shell. We notice that this shell encloses no mass. Unlike in case one where we enclosed the entire "planet". Since there is no mass this again implies:

$$\nabla \cdot \vec{g} = 0$$

Using the divergence theorem this yields:

$$\iint_{R_{inside}} \vec{g} \cdot d\vec{S} = 0$$

Again noticing spherical symmetry,

$$g(r) \cdot 4\pi r^2 = 0 \implies g(r) = 0$$

This again is exactly what we wanted. Which was to show that any point mass enclosed inside the sphere will experience no net gravitational force from the shell.

4 Applications

From our derivation we found that a point mass M outside the shell will experience a gravitational force of $g(r) = -\frac{GM}{r^2}$ We know that the force due to gravity on earth is $9.81\,m/s$. Here we attempt to find the force due to gravity on a planet similar to Earth, discovered by NASA in 2015, Kepler-452b.We know that the mass of Kepler-452b is approximately 3.29 times larger than earths, see [2]. Because we derived that gravitational fields outside a shell are equivalent to all the mass concentrated at the center, we can find gravity with just the mass and radius. So, we proceed by our formula:

$$g(r) = -\frac{GM}{r^2}$$

Using the approximated values:

- $G \approx 6.672 \times 10^{-11} \frac{Nm^2}{kg^2}$
- $M \approx 3.29(5.972 \times 10^{24}\,kg) \approx 1.965 \times 10^{25}\,kg$
- $r \approx 1.63(6378\,km) \approx 1.0369 \times 10^{7}\,m$

Using this we can deduce:

$$g(r) = \frac{(6.672 \times 10^{-11})(1.965 \times 10^{25})}{(1.0369 \times 10^{7})^2} \implies g(r) = 12.2\,m/s^2$$

This is approximately 1.24 times greater than Earths gravity which is $9.81\,m/s^2$

5 Conclusion

By using the divergence theorem to prove the Newton Shell Theorem we were able to simplify our calculations to find the gravitational force on a planet with just only the mass and radius. Using the divergence theorem is a lot easier than computing dense surface integrals. Not only can we apply this to Kepler-452b, but really any celestial object floating in space.

This is important because we can tie both mathematical ideas to physical phenomena. So in essence, we can use the divergence theorem to help predict gravity or maybe other phenomenon in space. Which is very powerful.All in all, the use of the Divergence theorem gives a clearer way to derive Newton's results in a much more precise and logical way.

References

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